

B30UE

Group 2

Ropeless Fishing Gear Demonstrator

Final Report

Dean Rowlett	David Valente	Ross Porteous	João Lopes
H00371472	H00347680	H00340210	H00342606
Dpr2000@hw.ac.uk	dv2000@hw.ac.uk	rp62@hw.ac.uk	jac2000@hw.ac.uk
Communications GitHub		Mobile Application GitHub	
Communications GitHub with Encryption			

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Abstract

This report details the development and validation of a ropeless fishing gear demonstrator designed to address marine entanglement – a major environmental issue – through the development of an on demand retrieval release mechanism. Developed at Heriot-Watt University, the project integrates mechanical, electronic, and software components to create a system that eliminates the need for vertical lines in fishing operations, thereby reducing the risk to marine life such as whales and sea turtles. The system utilises a Raspberry Pi Pico W, a Nanomodem v3.2 acoustic modem for underwater communication, and a Bluetooth-enabled mobile application for surface interaction. Extensive testing confirmed the system's functionality, demonstrating its capability to reliably activate and communicate between the seabed and surface units. Mechanical testing ensured durability and effectiveness in underwater conditions, particularly focusing on the magnet release mechanism which plays a critical role in the deployment of the buoy system. The mobile app developed allows fishermen to manage gear deployment remotely with ease, promising significant improvements in safety and efficiency. This report discusses the results of several field tests, the challenges encountered during development, and potential improvements for future iterations of the gear. The ultimate goal of this innovation is to foster safer, more sustainable fishing practices while mitigating human impact on marine ecosystems.

1 Introduction

Entanglement in fishing gear is a critical threat to marine life, such as whales and sea turtles, especially in regions like California and New England. In 2018 alone, the U.S. reported over 100 cases of entanglement, with 13 resulting in severe consequences, including injuries or death for the trapped animals. From work conducted during the September semester 2023 [1], these issues are addressed through the development of Ropeless Fishing Gear.

The initial phase revealed the adverse effects of traditional fishing methods on both marine life and fishermen, highlighting problems like rope entanglements, gear loss, and theft. This underlined the urgent need for a safer and more sustainable fishing solution.

Ropeless Fishing Gear offers a promising way forward, potentially transforming the industry by protecting marine species and improving the safety and efficiency of fishing operations. It aims to reduce gear loss and theft, benefiting both the environment and fishing communities.

Research Question:

How can the development of Ropeless Fishing Gear provide a sustainable solution to reduce marine life entanglement while improving the efficiency and safety of fishing practices?

To answer this, the project specification is focused on:

- Designing Ropeless Fishing Gear to eliminate the need for vertical lines.
- Implementing underwater communication for reliable gear release.
- Developing a user-friendly app for efficient gear management.
- Engineering durable components and electrical systems for the gear.
- Conducting extensive testing to ensure the gear's effectiveness and reliability.

This project seeks to address the pressing problem of marine entanglement with an innovative approach, marking a significant step towards sustainable fishing practices that better balance human activity with marine conservation.

2 Background Theory

The development of Ropeless Fishing Gear is grounded in extensive research and analysis of existing challenges and solutions within the fishing industry, particularly concerning marine life entanglement. The literature review and Preliminary Design Report from Semester 1 [1] has unveiled a range of innovative approaches aimed at mitigating the adverse effects of traditional fishing practices, paving the way for the conceptualisation and development of this project.

2.1 Existing Approaches and Technologies:

2.1.1 Sustainable Seas Technology, Inc. (Connecticut, USA) [2]:

- Approach: Implements a timed-release mechanism for its ropeless fishing gear. This system is designed to activate buoyancy controls at predetermined intervals, allowing fishing gear to surface for retrieval without continuous vertical lines.
- Impact: By employing a timer-based system, Sustainable Seas offers a straightforward solution to gear deployment and retrieval, minimising the window of entanglement risk. However, the predictability of timed releases might pose limitations in dynamic fishing operations.

2.1.2 Desert Star Systems LLC (California, USA) [3]:

- Approach: Utilises an acoustic signal for on-demand gear release. This method allows fishermen to remotely trigger gear retrieval, significantly reducing the likelihood of marine life entanglement.
- Impact: Desert Star's acoustic-based system aligns more closely with the project's vision, offering flexibility and responsiveness crucial for adapting to the unpredictable nature of marine environments and fishing operations.

2.1.3 Project Inspirations and Parameters:

Leveraging the operational insights from these companies, critical parameters were identified for the successful integration of the ropeless fishing gear solution:

These insights underscore the significance of adopting an acoustic signal-based activation method, similar to Desert Star Systems LLC's approach, for the project. This method offers the nuanced control necessary for responsive and adaptable fishing operations, directly addressing the research question and project objectives.

Parameter	Importance
Activation Method	On-demand, acoustic signal-based release provides flexibility and reduces entanglement risk.
Environmental Impact	Solutions must prioritise marine conservation while enabling effective fishing practices.
Reliability	Systems must perform consistently and reliably in diverse marine conditions.
Cost and Usability	Solutions should be economically viable and user-friendly to ensure wide adoption by fishermen.

Table 1 Project Parameters & Importance

The examination of Sustainable Seas Technology's timed-release system and Desert Star Systems LLC's acoustic signal activation contextualises the project. It reinforces the viability of acoustic signal-based solutions for ropeless fishing gear in achieving a sustainable balance between marine conservation and fishing efficacy. Inspired by these technologies, the ropeless fishing gear demonstrator is ready to make significant contributions towards safer, more sustainable fishing practices, emphasising the protection of marine life and the enhancement of fishermen's operations.

3 Methods, Design, and Implementation

3.1 System Overview

The system designed for the ropeless fishing gear solution combines aspects of software, electrical and mechanical engineering to create an intuitive solution that reduces the number of ropes present in marine environments. The system limits the time ropes stem from the sea floor to the surface by relying upon a two-stage communication process which is used to orchestrate the deployment and release of traps on the sea floor. This solution differs from typical trap deployment as the trap's associated buoy will sink along with it as opposed to remaining on the surface as a marker.

The communication process begins with a user selecting commands on a mobile app. The commands selected will be sent through Bluetooth Low Energy (BLE) communication to a surface level Nanomodem that will be positioned in the water. These commands will be relayed by the Nanomodem and sent as acoustic signals down through the water to a receiving Nanomodem on the sea floor. When the seabed Nanomodem receives this signal, it will trigger the release of a buoy attached to the deployed trap. The released buoy, along with its rope, will then rise back to the surface allowing fishermen to collect their traps. A summary of the communication process can be viewed in Figure 1 below.

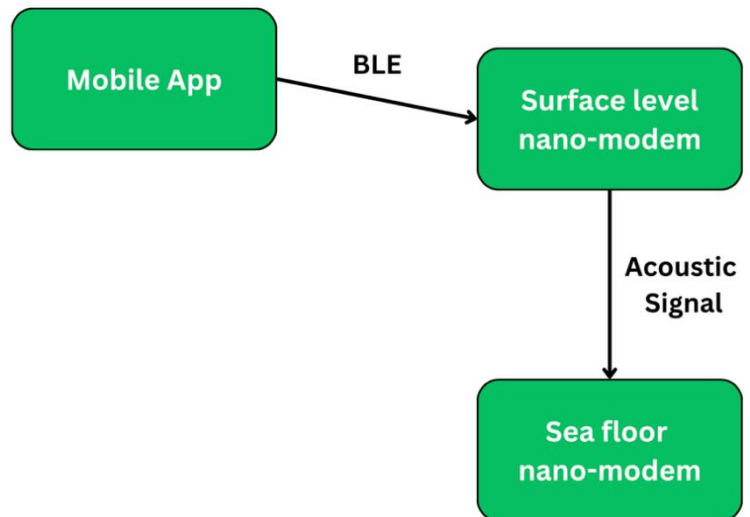


Figure 1 Communication process overview.

3.2 Electronic System Design

3.2.1 Previous work

A Heriot-Watt PCB was provided at the beginning of the project, developed from work carried out prior. After testing, it was clear that the PCB would not be suitable for the ropeless fishing gear demonstrator. This decision was several fold.

1) Inverted Rx and Tx Pinout

The board uses a Male DB9 connector for the RS232 interface with the Nanomodem. The pinout for this connector does not meet the standard RS232 pinout for male DB9 connectors, see Figure 2, with pin 02 & 03 reserved.

2) Incorrect Communication Protocol

An oscilloscope was used to interrogate each pin to find the Tx pin on the board. It became apparent that the idle voltage level for the Tx pin did not meet the expected logic level for RS232. The idle voltage for the Tx pin onboard the HWPCB was $\sim +6\text{V}$. The expected idle voltage level for RS232 Tx pin is within the range of $-13\text{V} \Leftrightarrow -3\text{V}$. This led to a further investigation into the pinout of previous and current boards.

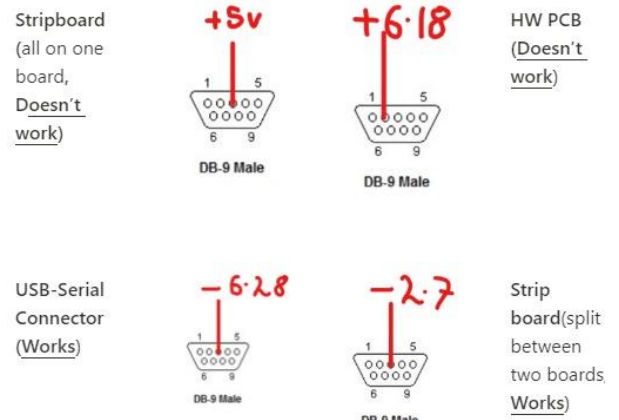


Figure 2 Tx Voltage Levels of available boards.

At this point of testing there were four methods available to interface with the Nanomodem. These included two stripboards from previous work, the Heriot-Watt PCB, and a USB-Serial RS232 Adapter. Of these methods two worked correctly, and the remaining two prevented the Nanomodem from starting correctly.

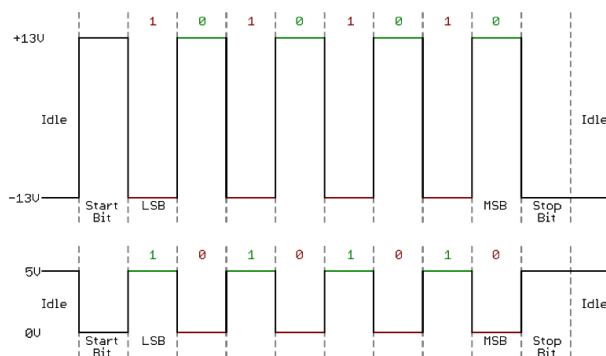


Figure 3 RS232 vs TTL protocol.

RS232 Voltages of ± 12 . Note how the -12V level corresponds to a logical '1' and $+12\text{V}$ level corresponds to a logical '0', which differs from the original TTL output - $+5\text{V}$ corresponds to a logical '1' and 0V corresponds to a logical '0', causing the Nanomodem to fail the startup sequence.

The final decision was that the failing boards did not use the correct protocol. After some research it became clear that these boards were using TTL protocol which has logic levels of $+5\text{V}$ for '1' and 0V for '0'.

Figure 3 shows a timed 0V & $+5\text{V}$ bit stream from the transmission pin of a microcontroller, such as a Raspberry Pi Pico W [4]. The upper diagram shows the translated signal levels at the RS232 driver output. These higher signal levels are generated by passing the $0\text{-}5\text{V}$ logic level to a transceiver chip, such as a MAX3232, which uses a diode/capacitor boost converter built in that boosts and inverts the signal levels to the required

3.2.2 Stripboard prototype

A provisional stripboard was created, see Figure 5, to amend the previous issues and allow for development of the other subsystems to continue. This stripboard uses a MAX3232 development board.

The stripboard follows the schematic seen in Figure 4. The wiring colours for the Nanomodem does not match pinout seen in the schematic. The wiring from the Nanomodem is seen in Table 2.

Nm3 Wire Colour	Wire	Schematic
Black	Vbat	Red
White	GND	Black
Red	TXD	Green
Green	RXD	Blue
Orange	RxS	N/A

Table 2 Nanomodem 3 Wire Colours

Several tests were carried out on the system using the stripboard as the main controller. However, due to the size of the DB9 connectors combined with the size restrictions within the enclosure, several repairs were required at the solder joints, primarily on the Nanomodem themselves. This caused several issues during the testing process and due to the project only utilising 2 of the available 8 pins on the connector (excluding ground pin), the decision was made to develop the PCB using a JST 4-way connector rather than the standard DB9 for the RS232 protocol.

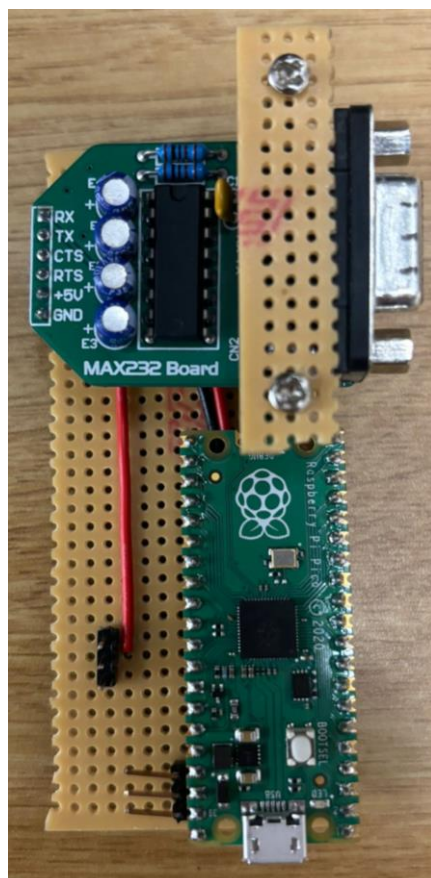


Figure 5 Provisional Stripboard

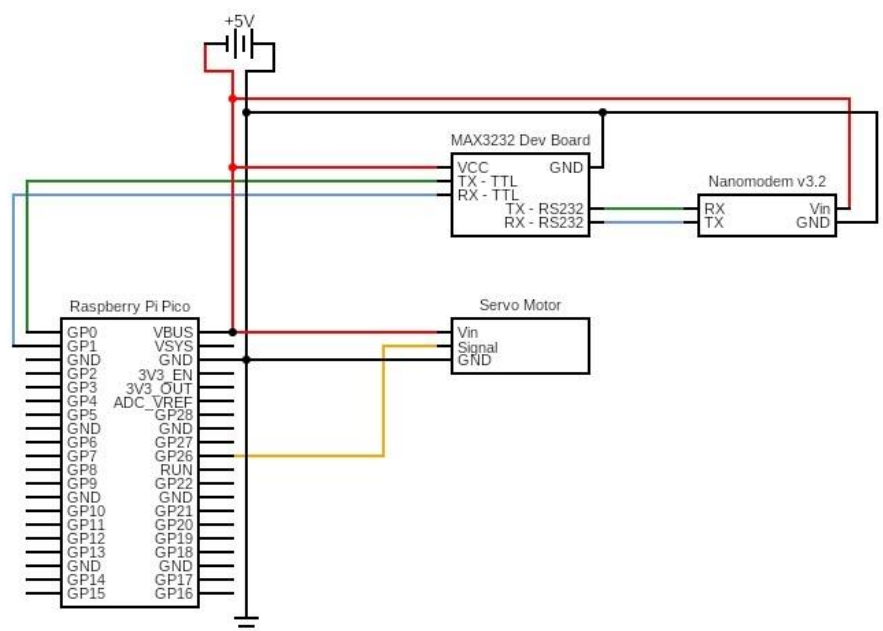


Figure 4 Stripboard Schematic

3.2.3 PCB Design

The PCB design for this system aimed to facilitate the control of the servo and the Nanomodem using a Raspberry Pi Pico W. This design was developed following the previous PCB manufactured at Heriot-Watt, for this project (Figure 6). This previous PCB was designed for interfacing with a stepper motor and a DB9 connector.

One of the essential components of the design is the MAX3232 transceiver, which has the role of translating the UART protocol used by the Pico to RS-232, compatible with the Nanomodem. While initially considering a MAX232, the MAX3232 was concluded to be more suitable for the intended purposes, given its compatibility with the 3.3V operation of the Pico.

In addition to the MAX3232, the PCB design featured a 3-pin JST connector dedicated to the control of servo. This connector served to provide power, ground, and PWM signals necessary for servo control. Furthermore, to interface with the nano modem, the initial plan was to use a DB9 connector. However, in an effort to optimise space on the PCB, the use of a 4-pin JST connector was opted. This decision was due to the fact that only four connections (power, ground, RxD, and TxD) were required to establish communication with the nano modem.

A crucial consideration during the design process was the constraint imposed by the electronics workshop's soldering capabilities. Specifically, the workshop was only equipped to make connections and solder components on the bottom side of the PCB. To accommodate this limitation, it was required to route as many traces as possible on the bottom side of the board. In cases where routing on the bottom side was not feasible, vias were strategically employed to connect top side traces back to the bottom side, ensuring accessibility for soldering.



Figure 6 Modified HW PCB

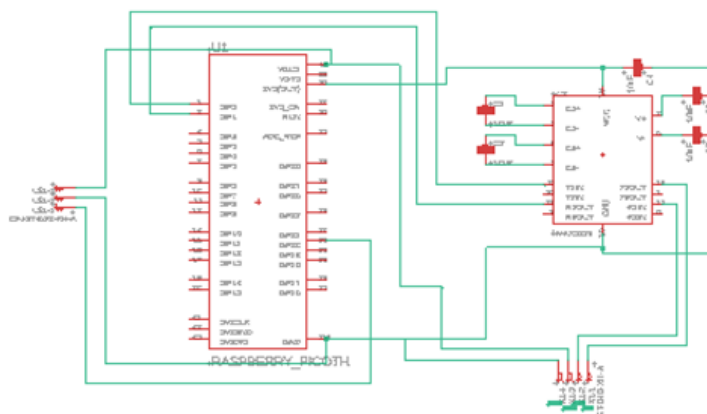


Figure 7 New PCB schematic.

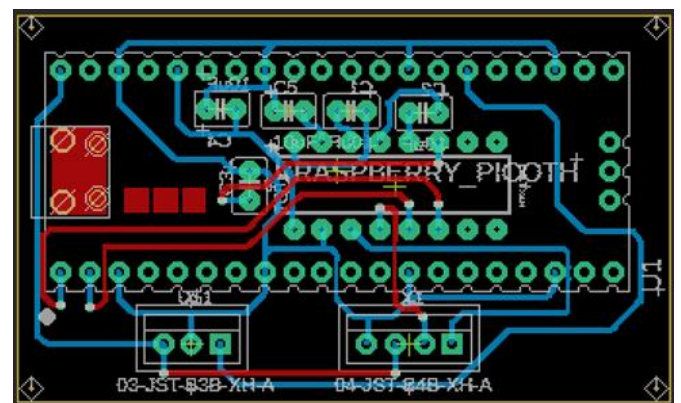


Figure 8 New PCB board design

Figure 7 shows the schematic of the new PCB and the connections between the Raspberry Pi Pico W and the JSTs and the MAX3232, and from the MAX3232 to the 4-pin JST. And Figure 8 illustrates the new PCB board design with the top layer traces in red and the bottom ones in blue.

Additionally, the electronics workshop modified some of the PCBs from last year. These PCBs contained similar components to the new design developed this year, but also included a stepper motor controller, which is not necessary for the new PCB since a stepper motor is not being used. With the collaboration of the workshop to remove the stepper controller from the existing PCBs, they were effectively repurposed for the project, while the new design was not complete. This meant that tests for other features of the system were able to be run.

In summary, the PCB design for this system was meticulously crafted to accommodate the specific requirements of this project, using components like the MAX3232 and JST connectors to facilitate the integration with the Raspberry Pi Pico W, the Nanomodem and the servo motor. Through careful

consideration of the necessary components and soldering constraints, the assembly and functionality efficiency were successfully optimised.

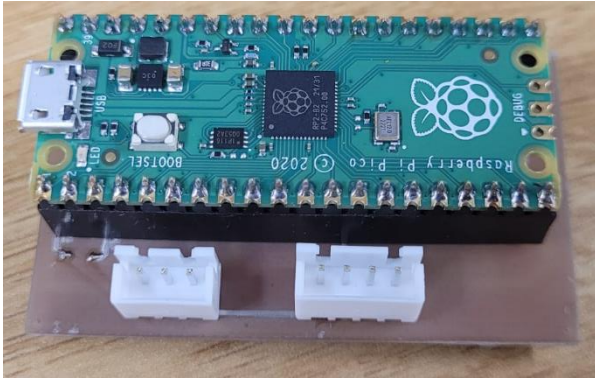


Figure 6 Top layer of new PCB

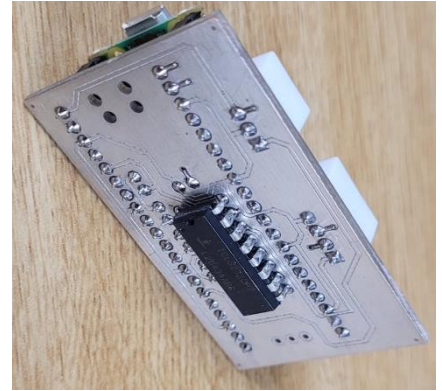


Figure 7 Bottom layer of new PCB

3.3 Communication System Design

3.3.1 Communication Protocols

The project used RS232 communication protocol. This protocol was dictated by the acoustic modem which exclusively used RS232. Throughout the project, a standard DB9 connection was used, see Figure 9, as an interface between the Nanomodem and the raspberry Pi Pico W. This was eventually changed in favour of a JST 4-way connector to minimise the overall footprint of the electronics.

The communication protocol was set up over a serial port with the following configured settings; a baud rate of 9600, 8 data bits, 1 stop bit and no Parity, as described in the Nanomodem v3.2 datasheet.

3.3.2 Communication System Architecture

The system consists of two main elements: a seabed unit and a surface unit, both powered by Raspberry Pi Pico W controllers. Both units use a Nanomodem v3.2 from Newcastle University, for underwater signal exchange. The systems operate through MicroPython scripts:

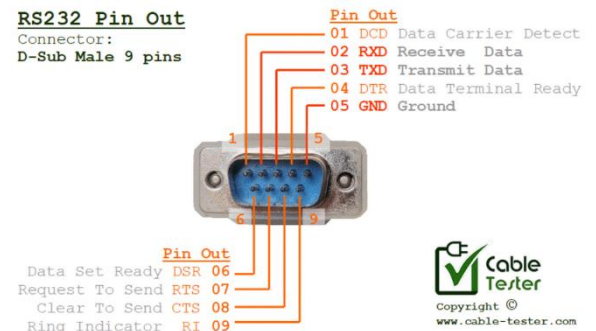


Figure 9 Standard Male DB9 - RS232 Connector

nm3_pico_driver.py

This serves as the communication driver with the acoustic modem, facilitating signal sending and receiving.

Servo.py – seabed unit

This handles the Micropython servo library and contains a user-friendly function for changing the servo motors position.

main.py – seabed unit

This script - uploaded to the seabed unit - waits for a release command from the surface, using the servo to release the buoy once the command is received.

main.py – surface unit

This script – uploaded to the surface unit – manages Bluetooth communication with the user's app, enabling command transmission to the seabed unit.

3.3.3 Surface Unit Design

The software design for the surface unit follows an asynchronous design.

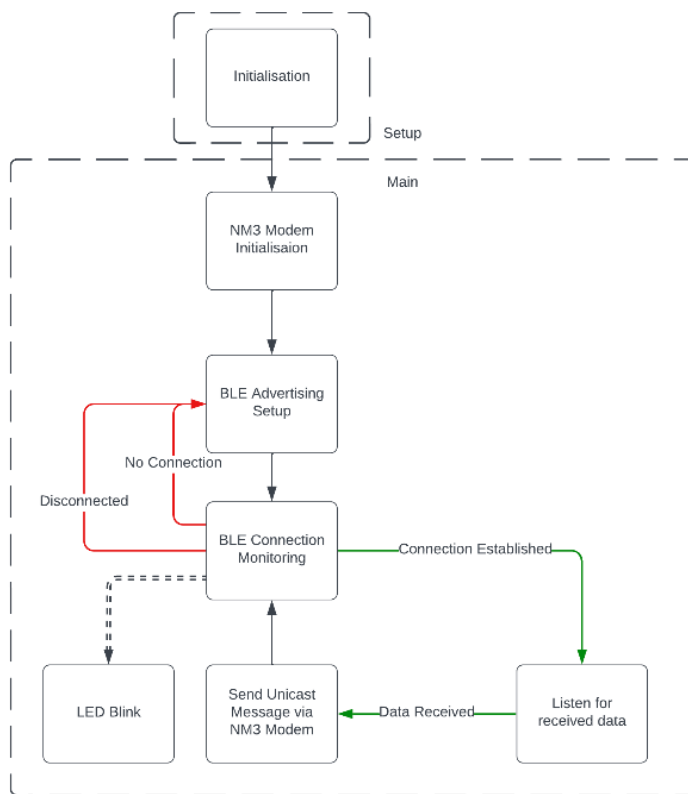


Figure 10 Surface unit Software Diagram

3.3.4 BLE (Bluetooth Low Energy) Implementation

The Bluetooth Low Energy (BLE) communication protocol provides fast and reliable communication between the mobile device and surface level Nanomodem. It was chosen in this project primarily due to its low power consumption relative to other communication protocols such as Wi-Fi or Classic Bluetooth. This was a key requirement for the project as the Nanomodem may have infrequent charging opportunities if regularly used out at sea.

In addition, BLE offers a lower latency than Classic Bluetooth which allows units of data to be transmitted to the above surface Nanomodem faster. This will allow for faster communication between devices which ultimately results in a smoother user experience. In the past, BLE may not have been a great option as the technology was still relatively new to smart phones and was not a common feature to inexpensive microcontrollers. However, this has now changed as BLE has been adopted by many more electronic devices and is a ubiquitous feature of modern smartphones.

A secure BLE connection must be established between devices before any communication is attempted. BLE devices have three primary operating modes to facilitate the connection between two devices. These are advertising, scanning and connected. BLE also requires that devices be setup using services which are used to group related functionality and the commands that will be sent between devices. Each service is defined with its own Universally Unique Identifier (UUID) which is a 128-bit label used to uniquely identify the services. Services can be setup specifically to identify devices for connection.

When the mobile app is first loaded, it will automatically begin scanning for nearby devices that contain a service with a UUID that matches its own. Conversely, the surface level Nanomodem will by default always be advertising with a service that contains a UUID matching the one on the app. Once a match is found, the two devices will transition from their respective scanning and advertising states to connected. Data can now

be sent between the two devices enabling a user to write commands from the app to the surface level Nanomodem. The type of commands that can be sent from the mobile app will be explained in more detail in section 3.5.

3.3.5 Seabed Unit Design

The seabed unit is an embedded system designed to operate attached to a trap on the ocean floor. It is equipped with a Raspberry Pi Pico W, an NM3 acoustic modem for underwater communication, and a servo motor mechanism to manage the release of the buoy. Upon receiving a specific acoustic signal, the unit triggers the servo to alter the state of a magnet, either releasing or securing the buoy based on the command. The system also features energy-saving measures, entering deep sleep mode to conserve battery life during inactivity and waking at predefined intervals to check for new commands.

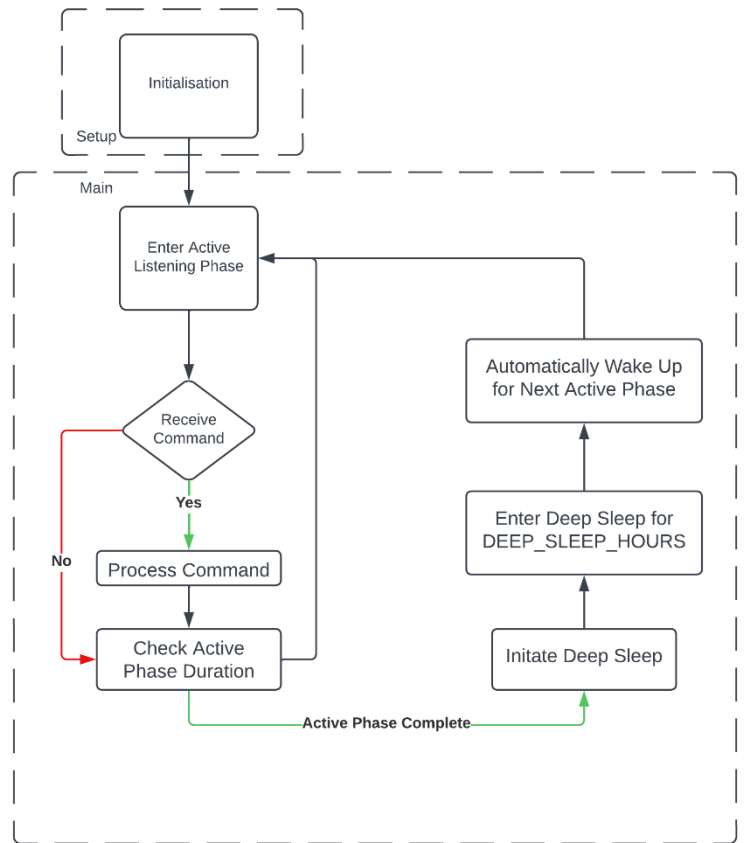


Figure 11 Seabed unit Software Diagram

3.3.6 Nanomodem v3.2

Acoustic communication is handled by the Nanomodem v3.2 (Nm3). This modem uses serial communication with certain setup parameters, see Communication Protocols. The modem prefixes each issued command with '\$' and does not require terminating characters. Sent commands are acknowledged locally with a response prefixed with '\$' or a response from the target modem prefixed with '#' and terminated with <CF><LF>.

The table below shows the transmission timing for all command types available on the Nanomodem v3.2.

Command Type	Serial Data transmission timing at 9600 baud (seconds)
Ping request/response	0.105
Command message (\$Vxxx, \$Txxx)	0.105
Data message (n bytes)	$0.105 + (n+16) * 0.0125$

Table 3 Nanomodem Serial Data Transmission timings

Message and command packets sent from the Nm3 are formatted in a specific structure, see Figure 12.

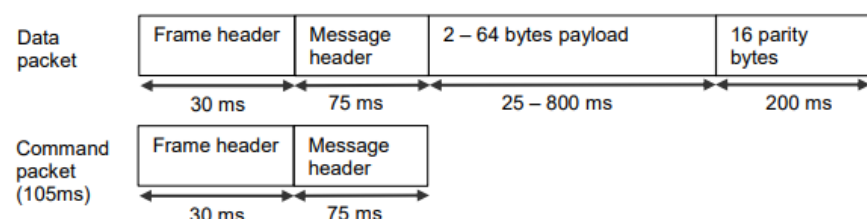


Figure 12 Nanomodem Data and Command Packet Structure

Figure 12 contextualises the data and command packet timings shown in Table 3. Command packets are interpreted by a prefix character. Data transfer commands accept up to 64 bytes with each byte transmission adding 12.5ms, making their max transmission timing 1.105s.

The project utilises the send unicast data function of the Nanomodem. Unicast Data has the following structure:

\$Uxxxnnddd...	
\$U	Message identifier
xxx	Address of target address (0-255)
nn	2-digit decimal representing the message length
ddd	Data to be sent, as bytes

Table 4 Send Unicast Data structure.

Given the above message structure for sending data between two modems, a driver was written to incorporate this in a user-friendly manor.

The code shown in Figure 13 shows the driver implementation of sending unicast data. Where the send command implementation method is seen in Figure 14.

```
def send_unicast_message(self, address, message):
    message_length = len(message)
    command = f"$U{address:03d}{message_length:02d}{message}"
    self.send_command(command)
    debug_print(f"Sending Command: {command}")
    return self.read_response()
```

Figure 13 Send Unicast Message Method

The send_command() method takes the string argument which encodes the string with utf-8 (Unicode Transformation Format – 8 bits) structure and writes this encoded string to the UART port specified during setup, in this example the UART is set up on bus 0.

```
def send_command(self, command):
    self.uart.write(command.encode('utf-8'))
```

Figure 14 Send Command Method

This encoding of this command can be seen as bytes and UTF-8 in Table 5.

String	Bytes	UTF-8
\$U	24 55	\u+0024 \u+0055
16007	31 36 30 30 37	\u+0031 \u+0036 \u+0030 \u+0030 \u+0037
RELEASE	52 45 4c 45 41 53 45	\u+0052 \u+0045 \u+004c \u+0045 \u+0041\u+0053 \u+0045
U\$16007RELEASE	24 55 31 36 30 30 37 52 45 4c 45 41 53 45	\u+0024 \u+0055 \u+0031 \u+0036 \u+0030 \u+0030 \u+0037 \u+0052 \u+0045 \u+004c \u+0045 \u+0041 \u+0053 \u+0045

Table 5 Release Command Encoded

The timing of sending a unicast message can be seen below, see Table 6.

Message structure	Message input	timing
Frame Header	\$U	30ms
Message Header	16007	75ms
Bytes Payload	RELEASE	7 * 12.5ms = 87.5ms
16 Parity Bits		16 * 12.5ms = 200ms
Send Unicast Data Message example	\$U16007RELEASE	392.5ms

Table 6 Release Command Timings

Additionally, a method was developed to integrate the distance between two modems, see *Figure 15*. This feature utilised the ping feature of the Nm3 which returns the time taken to send a command message between two modems.

The method sends a ping command to the specified modem address, waits for a local acknowledgement then, once this is received, the response is read again with the timing response removed and assigned to the *raw_distance* variable. The following formula is then applied to this time to get the distance:

$$distance = raw_distance * c * 3.125e-5$$

where *c* can be changed between two constants representing the speed of sound in air and water, depending on the testing environment.

```
def ping(self, address):
    water = 1500 # 1500m/s in water
    air = 340 # 340m/s in air
    command = f"$P{address:03d}"
    self.send_command(command)

    # Wait for command acknowledgment
    ack = self.read_response()
    if ack is None or not ack.startswith('$P'):
        raise Exception("No acknowledgment received")

    # Wait for range response or timeout
    response = self.read_response()
    if response is None:
        raise Exception("No response received")

    if response.startswith('#R'):
        raw_distance = int(response.split("T")[1])
        sound_velocity = water
        c = 0.00003125 # Conversion factor
        distance = raw_distance * sound_velocity * c
        return distance
    elif response.startswith('#TO'):
        raise TimeoutError("Response Timeout")
    else:
        raise Exception("Unexpected Response")
```

Figure 15 Ping Method

3.4 Mechanical System

The mechanical design of the Ropeless Fishing Gear prototype is fundamental to its functionality and effectiveness in mitigating the entanglement of marine life in fishing gear. This section outlines the methods, design considerations, and implementation steps undertaken to develop a robust and reliable mechanical system.

3.4.1 Initial concept drawing

Before starting to build any system, it is important to first plan and sketch the full system (Figure 16). As the system progresses, the drawing can be recursively refined and amended as necessary until the project is complete.

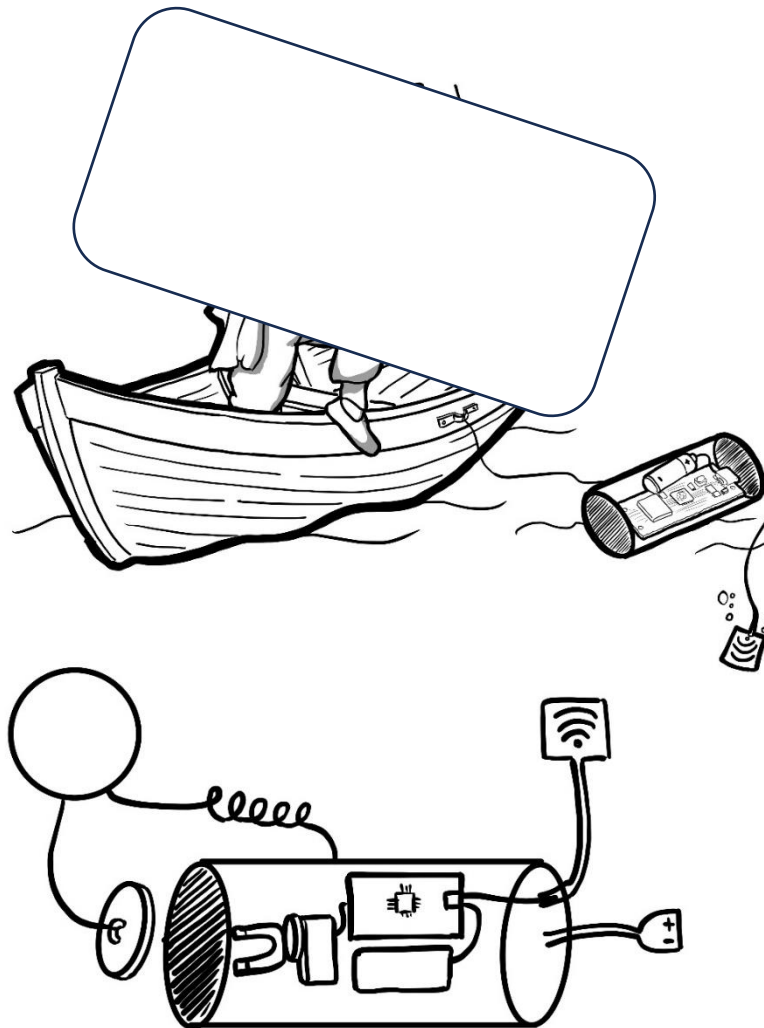


Figure 16 Initial Concept drawing

3.4.2 Waterproofing

When designing a system which is going in the sea, waterproofing is one of the top priorities. If the system is not fully waterproof up to a desired depth according to the specification, then the product will inevitably fail. This section of the report details the different measures taken to ensure that the product is fully waterproof and how this claim was tested.

The parts of the system responsible for keeping it watertight and their functions are as follows:

- The flange (Figure 17) is a piece of metal used to make the connection between the acrylic enclosure and the endcaps, it features three grooves where O-rings are inserted to ensure a tight seal.
- Penetrators (Figure 19) are attached to the endcap and their function is to waterproof the connection between electrical cables when going through the endcap. They also include O-rings for tight waterproof seal. To further reinforce the connection between the electrical cables and the



Figure 17 75mm Flange from BlueRobotics



Figure 19 6.5mm penetrator from BlueRobotics



Figure 18 Araldite potting compound

penetrators, a potting compound was also used. Araldite (Figure 18) is a strong lasting adhesive which is water resistant and can bond to the materials used. Other advantages include fast setting times and being environmentally friendly.

- After consulting the BlueRobotics website, the use of a product called Molykote 111 (Figure 20) was recommended. Molykote 111 is a type of silicone grease used to lubricate O-rings and other seals. It was used in all the seals and around all the O-rings throughout the enclosure.



Figure 21 Silica gel from RS PRO

- After making sure all the above had been implemented, bags of silica gel desiccant were also added to each of the enclosures. They absorb ambient moisture and prevent corrosion. RS PRO silica gel desiccant (Figure 21) comes in the form of hard irregular shaped crystals. The material is designed to absorb moisture from air, gases, and liquids to prevent the relative humidity (RH) rising to a point when corrosion, fungal growth and optical smearing will occur.



Figure 20 Molykote 111 compound

3.4.3 Assembly and materials used

3.4.3.1 Pressure vessel

When deciding on the pressure vessel, a few important factors had to be taken considered, namely the size of the enclosure, the material, brand, number of parts available to fit the enclosure and finally the depth rating.

After carefully consideration, it was decided that the best option would be to use an acrylic enclosure measuring 225mm in length, 75mm inside diameter, with a depth rating of 150m supplied from the BlueRobotics website. Purchasing an enclosure from BlueRobotics ensures a certain amount of quality as it comes from one of the most reputable brands in the underwater robotics business. BlueRobotics also offers an array of parts for their acrylic enclosures which guarantee that the enclosure will stay watertight.

3.4.3.2 Magnet

A magnetic switchable device (sometimes called magnetic base) is a magnetic fixture that uses one or more permanents magnets in a configuration that allows the external field to be turned on or off.

One type of magnetic switchable device is made from two blocks of iron, with a round cavity bored through the centre. The halves are joined together with a non-ferrous material such as brass or aluminium. A round permanent magnet is inserted into the bored hole and a handle is attached to allow rotation of the magnet (Figure 23). This act of rotation changes the orientation of the magnetic field.

In the “off” position (Figure 22), the poles are oriented towards the non-ferrous core. The iron blocks act as keepers by bridging between both poles.

In the “on” position (Figure 24), the poles are each in one iron half, which then acts as an extension. The field is effectively passing across an air gap (at the base and top). If this gap is bridged with a piece of iron, it becomes part of the magnetic circuit and will be attracted with the full strength of the magnet.

A magnetic base can therefore be attached in a variety of positions to any ferrous surface, allowing the base to be positioned in the desired orientation.

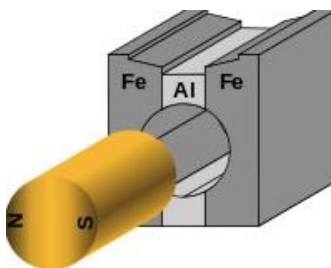


Figure 23 Base and magnet assembly (inverted position)

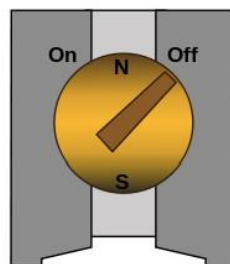


Figure 22 Magnet in the "off" position

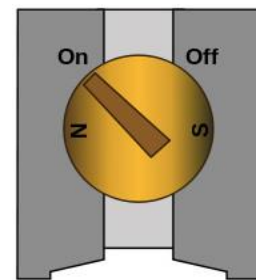


Figure 24 Magnet in the "on" position

3.4.3.3 Stainless steel disc

Since stainless steel was the material which best conducted the magnetic fields through different endcaps, it was decided that stainless steel would also be the best material for the manufacture of the disc. The purpose of the disc is to have an anchor point so that the rope leading up to the buoy can be attached.

Since the dimensions of the base of the magnet are in the shape of a 50mm square, the size of the disc must be similar to this and must not exceed the distance of 78mm between the bolts securing the endcap in place as shown in Figure 26.

Therefore, it was decided that an outer diameter of 60mm for the stainless steel disc would be an appropriate size as it was just over the size of the magnet base, while staying flush against the endcap since it did not exceed the distance between the bolts. Additionally, by having the outer diameter of the disc slightly smaller than the distance between the bolts, this ensured that the disc did not slide off the sides as the bolts lock the disc in place sideways.

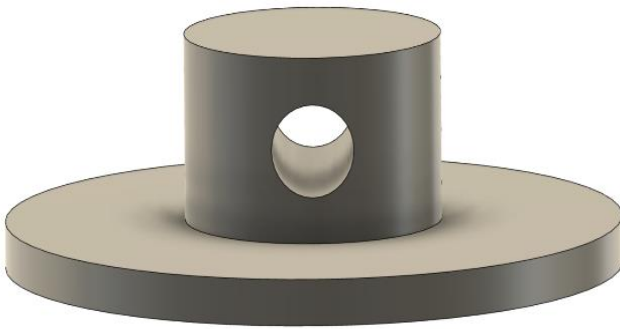


Figure 25 3D model of disc

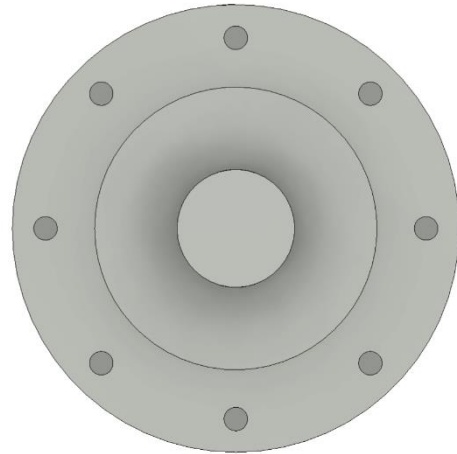


Figure 26 Front view of disc, endcap and bolts

After defining the criteria for a successful disc, a simple CAD model was produced using Fusion360 (Figure 25) and any final dimensions were adjusted.

Finally, the design was given to Heriot-Watt's mechanical workshop and within one week, the stainless steel disc was produced and later tested (Figure 27)



Figure 27 Manufactured stainless steel disc

3.4.4 CAD model

A computer aided design (CAD) shown in Figure 28 was produced using Fusion 360. It gives a good indication of the final design, what components it consisted of and how they all fit together inside the enclosure. The size of the 3D model is to scale with the design of the physical model. Since each component can be set to “visible” or “hidden”, it is possible to isolate and check the specific dimension of each individual component. By having it saved online, it can be easily shared and viewed by anyone from anywhere in the world.

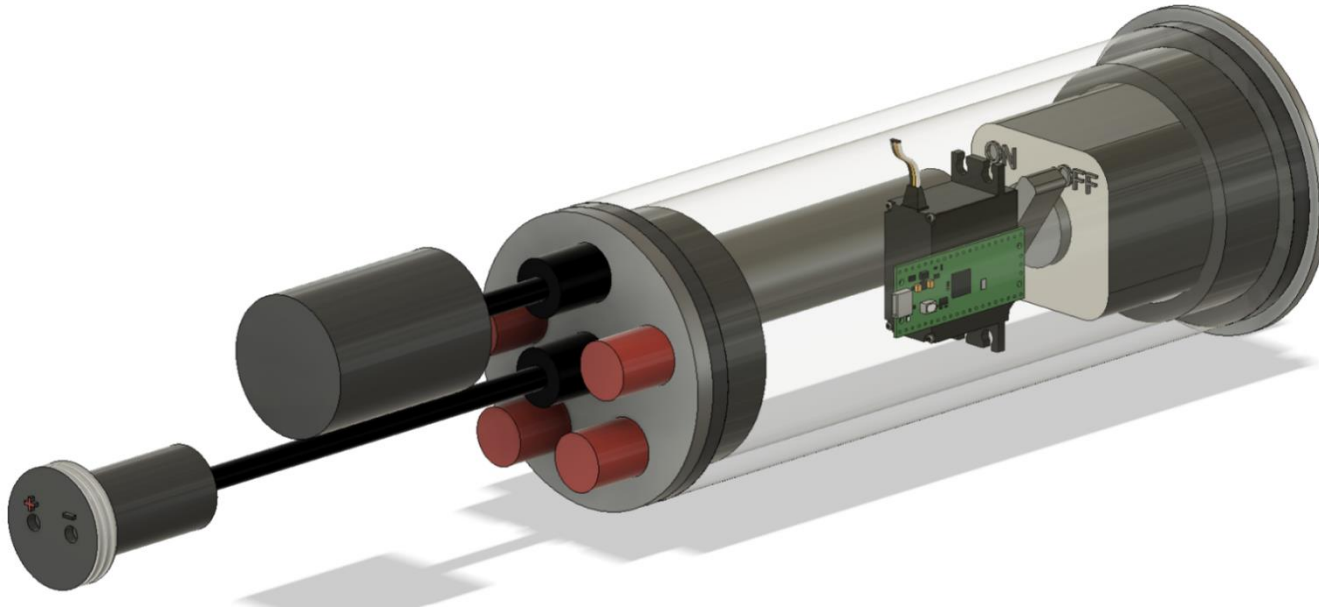


Figure 28 CAD model.

3.5 Mobile App

3.5.1 Overview

The ropeless fishing gear app combines an intuitive user interface with powerful capabilities enabling fishermen to easily deploy and manage their traps. It serves as the first step in a chain of operations, beginning with the press of a button on the app and culminating with the release of buoy on the sea floor. It performs this step by sending commands to the surface level Nanomodem via Bluetooth Low Energy (BLE) communication. The app was built using the JavaScript and Python programming languages and it utilised the React Native cross-platform development framework.

3.5.2 Development Plan

The app was implemented over various phases and many of its later additions built upon previous features. When implementing the app, key objectives were identified and prioritised at the beginning of the project. These included designing an attractive user interface and establishing a reliable Bluetooth connection with the Nanomodem. The initial plan used for building the app can be viewed in Table 7 below.

Phase	Task	Duration
1	Familiarise with React Native capabilities	Winter Break
2	Design app layout and complete wireframes	Weeks 1 - 2
3	Implement the layouts of key screens	Week 3
4	Implement Bluetooth capabilities on the app	Weeks 4 - 6
5	Setup Raspberry Pi Pico to receive data via Bluetooth	Week 7
6	Streamline user set-up process	Week 8
7	Add a feature to request and save a user's current location	Week 9
8	Integrate app with the Nanomodem	Week 10
9	Add a help and guidance page to the app	Week 11
10	Test end-to-end process	Week 12

Table 7 Mobile app development plan.

3.5.3 Tools and Technologies

3.5.3.1 *Mobile App*

One of the main requirements of the app was for it to be accessible to as many users as possible. This was to encourage more people to adopt the ropeless fishing solution and ultimately reduce the impact ropes have on local wildlife. Taking this into consideration, it was decided that the app would be built for both iOS and Android devices which would cover the two major mobile operating systems.

There are two common approaches to mobile app development with each offering its own unique benefits. The first and more traditional approach is native app development which is a methodology that exploits development frameworks and tools designed for a singular mobile platform. This approach allows a developer full access to the features of one specific platform, but it does not allow the app to be published to both iOS and Android. The second approach is cross-platform development, where an app can be built once using common tools and then published to both iOS and Android platforms.

The ropeless fishing gear app was built using cross-platform development as some of the more advanced features unique to each respective platform were not needed for this application. The cross-platform approach also helped keep costs and development time low as the app could be built once and later optimised for each platform. There were several tools identified for cross-platform mobile development which included Flutter, React Native and Kotlin. For this project, the React Native framework was selected due to its design functionality and clear documentation. React Native also offers support for Expo, a cloud platform that provides remote servers to run mobile simulations and tests. This was extremely useful throughout development and during debugging as the app could be viewed on physical mobile devices which gave access to the Bluetooth hardware necessary for testing.

3.5.3.2 *Bluetooth Receiver*

The surface level modem was powered by the Raspberry Pi Pico W microcontroller. In relation to the mobile app, it was responsible for receiving and interpreting BLE messages sent from the app. The microcontroller was programmed in MicroPython, and it was developed using the Thonny integrated development environment.

3.5.3.3 *Version Control*

Both the mobile app and Bluetooth receiver had their own code repository set up using GitHub version control. This provided a safeguard as it would make it easy to roll back to previous app versions in the event that any issues were created when new features were added. Sharing the centralised GitHub repository also encouraged collaboration with other team members when separate components of the project were integrated.

3.5.3.4 *Development Server*

A dedicated development server was required in order to test and simulate the mobile app on physical devices. Initially, a simple local server was used on a private Wi-Fi network which allowed physical mobile devices to download and use the app. This local server worked fine on private Wi-Fi networks, but it presented security problems when ran on a public Wi-Fi network. Due to this, the firewall of many public Wi-Fi networks would automatically block local servers which prevented the mobile app from being ran on a physical device.

Addressing this, Expo Go's cloud computing environment was used to run remote servers that hosted the app. This allowed physical mobile devices to connect to the same remote server and download the mobile app which provided a suitable testing solution.

In the later stages of the project when testing the app in the lab environment became crucial, a dedicated private Wi-Fi network was setup and accessed for the final local tests.

3.5.4 App Functionality

3.5.4.1 Navigation and Key Components

The app aims to assist fishermen with the management and deployment of their traps. It abstracts the complexities of BLE communication and streamlines the trap set-up process enabling users to get started in a short time without any hassle. To enable this complex functionality, the app consists of five main pages that all play an essential role. The main pages in the app include the control panel, add page, trap page, help page and settings page. Figure 29 outlines each of the key pages and displays how users are able to navigate between them.

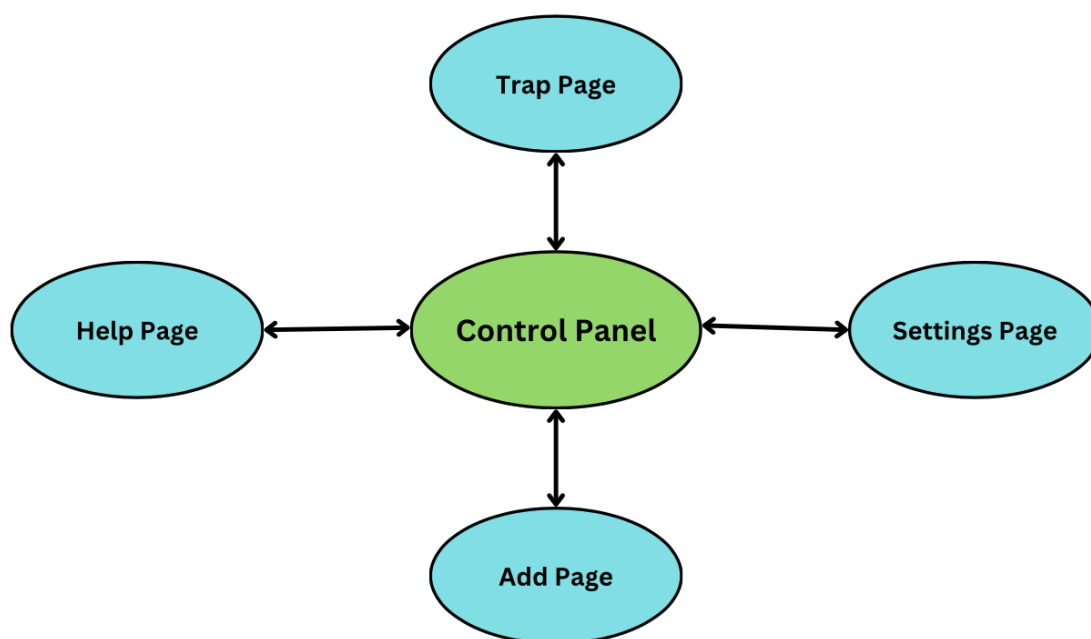


Figure 29 Relationships between pages within the app.

3.5.4.2 Control Panel

The control panel serves as one of the core components of the app and it is the first screen users will be met with when it is launched. This screen provides users with an overview of all of their traps, and it simplifies the navigation to other sections. An example of this screen can be viewed in Figure 30 below.

The control panel is designed to be easy and efficient to use for people out at sea. It follows a minimalistic design and features larger buttons to avoid pernicky control and unintended commands. The control panel also features a customisable user interface as users have the options to switch between grid and list view using the toggle at the top left of the screen. In Figure 30, the screen on the left highlights the grid view and the screen on the right displays the list view option.

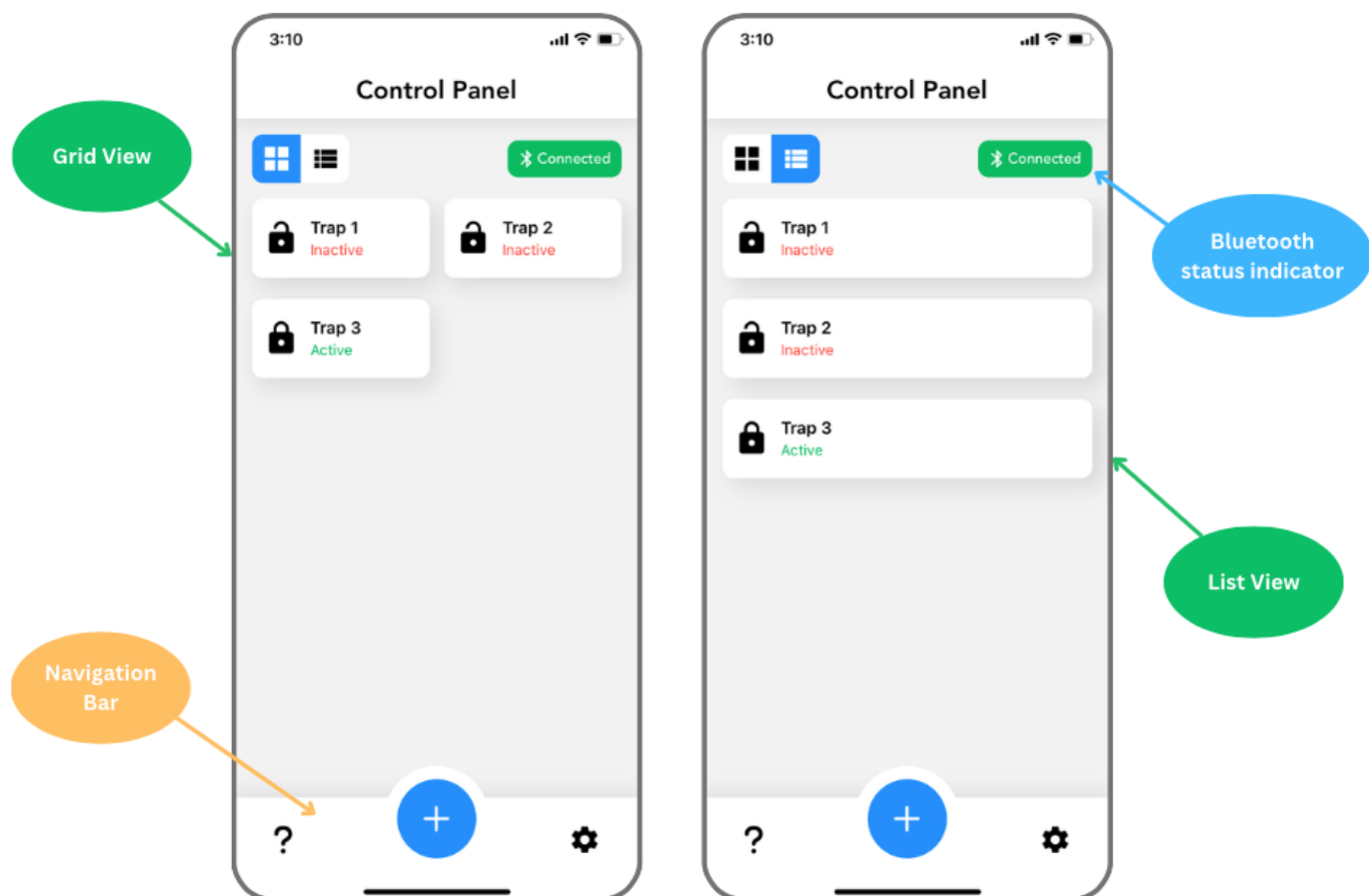


Figure 30 Control Panel examples.

3.5.4.3 Add Page

The add page allows users to create and add new traps to their account. It utilises a simple form that makes it easy for users to enter the required information to identify and setup new traps. An example of this screen can be viewed in Figure 31 below. The screen on the left shows the page before any data has been entered and the screen on the right displays the page after a user has entered the data for their trap.

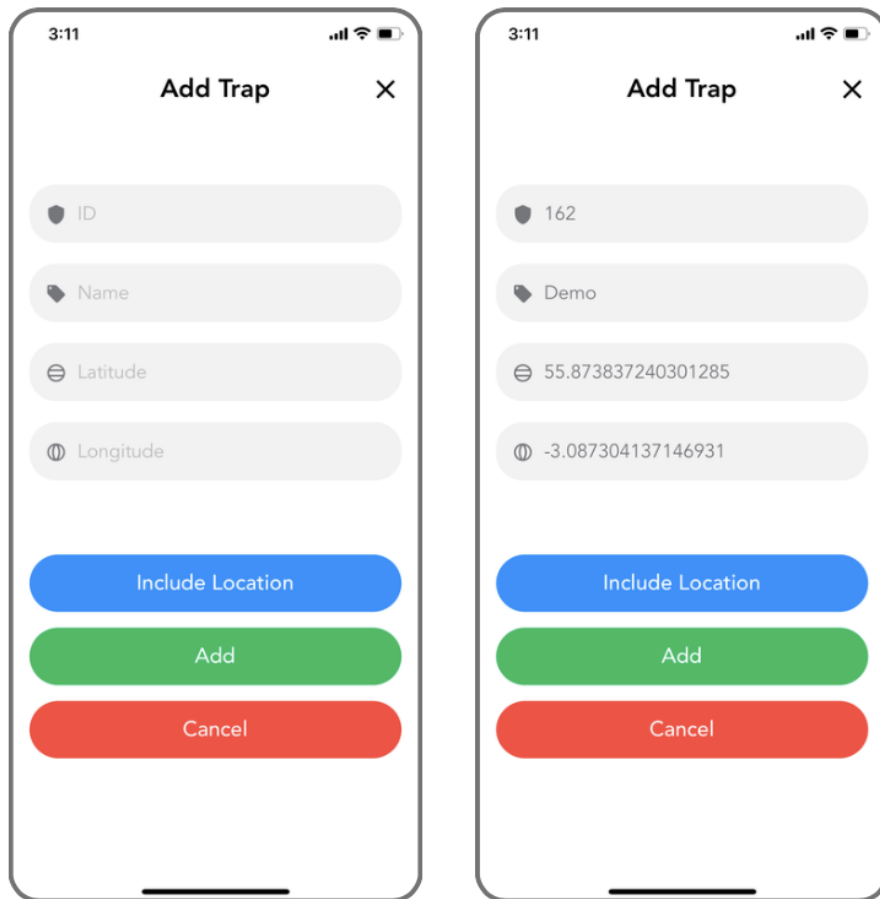


Figure 31 Add page examples.

One issue encountered during development was how traps would be uniquely identified and linked as the number of traps scaled. Due to this, it was crucial to employ a method to uniquely identify each trap as without this, commands would likely be sent to incorrect traps if they shared a common ID. Addressing this, the add page requires users to enter a unique ID for each new trap by preventing new traps using duplicate IDs from being added.

In addition to this, each of the above surface Nanomodems were assigned unique IDs which matched their address. These IDs were labelled on the side of the modems. When adding a new trap, users should use the ID present on the side of their Nanomodem as this will prevent any duplicate IDs from being added and it will assist with the communication between the mobile app and Nanomodems. It is important for a user to use the ID displayed on their Nanomodem as this matches the address of the Nanomodem the app will send commands to. Guidance on this setup process is provided in the help page which will be explained in more detail in section 3.5.4.5.

Extending on this, the add page provides users with the option to record their current location by providing latitude and longitude values. This can be done by manually entering the values or by selecting the 'include location' button. This button will automatically populate the above latitude / longitude fields by requesting a user's location using the Global Positioning System (GPS) functionality built into most modern smartphones.

To finish this setup process, a user can simply select the 'add' button which will confirm the operation and add the new trap to their account. Alternatively, if the new trap is no longer required, a user can select 'cancel' which will stop the operation and return them to the control panel.

3.5.4.4 Trap Page

Once users have added a new trap, they will be able to select this from their control panel which will bring them to the 'trap page'. This page allows users to easily set and release their traps by selecting the large circular button in the centre of the page. The icon on this button depends on the status of the trap and it provides users with an indication of the operation they are about to perform. If the status is inactive, the icon will display a closed lock which indicates that the trap will be set upon pressing the button. If the trap is active, the icon will display an open lock, indicating that the user is about to release the trap. An example of this page can be viewed in Figure 32 below.

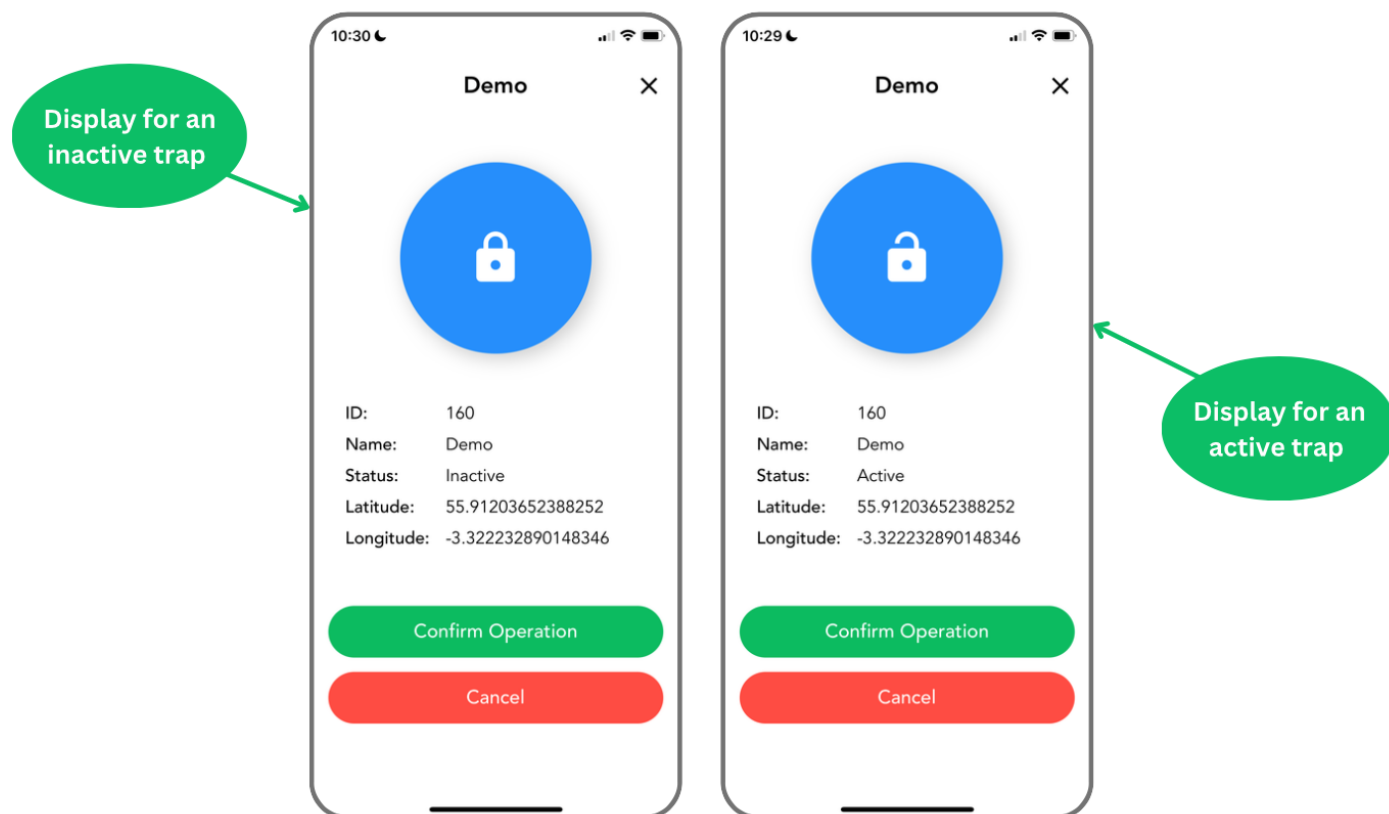


Figure 32 Trap page examples.

To avoid any unintended commands being sent to the Nanomodem when the main button is selected, the user will be alerted with a warning asking them to confirm the operation before the command is sent. The warning they receive includes the name of the trap and the type of operation being performed. For example, if a user is attempting to release 'Trap 2', they will be alerted with the following warning: "Are you sure you want to release Trap 2?". If they select 'YES', the operation will be performed and conversely it will be cancelled if they select 'NO'. This warning provides an extra layer of protection against accidental button presses or incorrect operations.

The trap page also conveniently displays the data relevant to the selected trap such as the ID, name, status, latitude, and longitude. This provides a quick summary of key information and having easy access to view the latitude and longitude values could assist fishermen locate their traps.

3.5.4.5 Help Page

One of the main objectives of the mobile app was it for it be easy to use and accessible to those who may not be experienced with technology. In response to this, efforts were taken to streamline the setup process

and abstract as much of the complexities of BLE communication as possible. However, when designing a mobile app that interacts with the real world, some form of setup process and cooperation from users is inevitable. To further assist users with the setup process, a help page was added to the app which aims to make the setup and usage of the app simple and pain free. An example of the help page can be viewed in Figure 33 below.

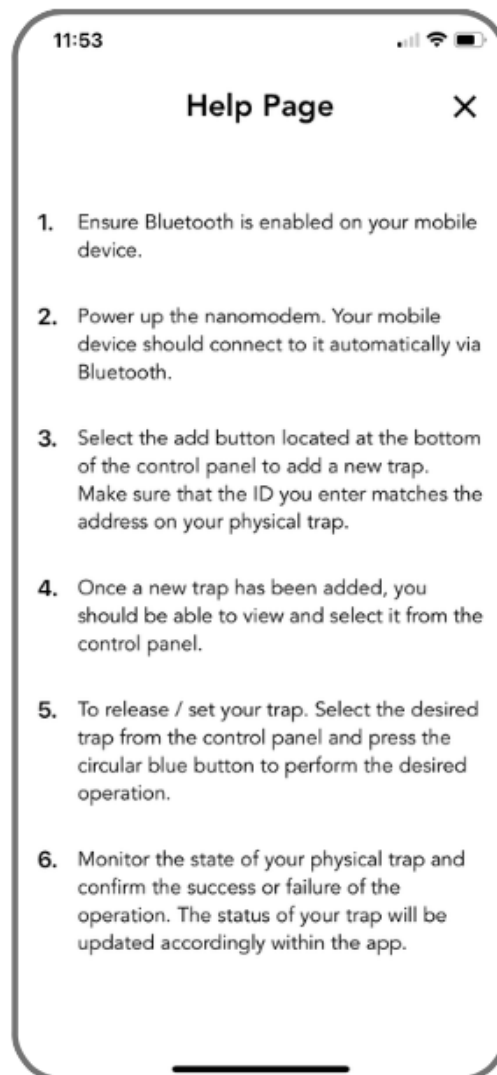


Figure 33 Help page example.

3.5.5 Bluetooth Low Energy

The mobile app communicates with the surface level Nanomodem via Bluetooth Low Energy (BLE) communication. The connection process is abstracted from users as the mobile app will automatically scan and connect to Nanomodems within its vicinity as long as a user's phone has Bluetooth enabled. In the event of an unexpected power outage or disconnection, the mobile app will automatically begin trying to reconnect to the Nanomodem until a secure connection is once again established.

The status of the BLE connection is displayed on the top right of the app control panel and it can be one of four values. These are scanning, connecting, connected and reconnecting. The transition between these states is summarised in Figure 29 below.

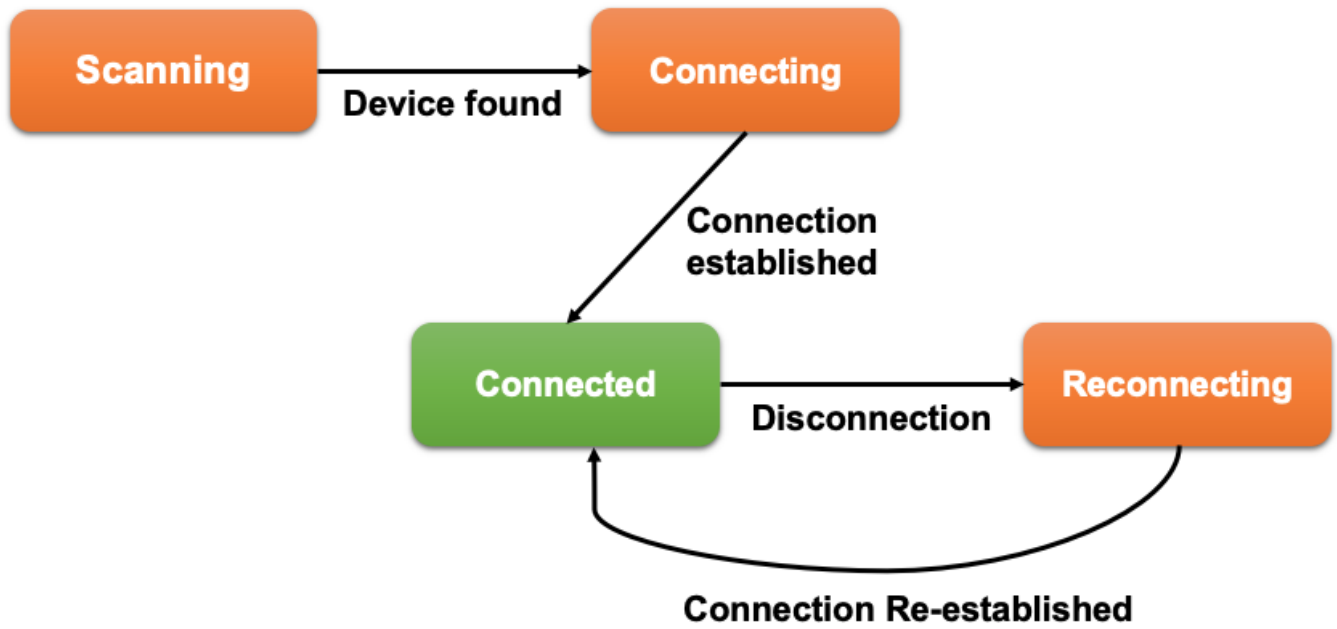


Figure 34 BLE status transition diagram.

4 Results, Verification, Validation

The results obtained during the various tests carried out throughout the project can be seen below. Overall, the testing was strong, with a decreased success rate during testing at the Heriot-Watt Loch. Due to the shallow nature of the loch, the testing suffered from severe multipath errors, with the acoustic signals reflecting consistently from the water surface. However, testing carried out vertically within the Ocean Systems Lab obtained improved results, further demonstrating the effects of multipath errors. Testing out with water obtained consistently good results but have a maximum range of ~2m.

4.1 Results Table						
Location	Test Medium	Description	Success (%)	Range (meters)	Date	Notes
Ping testing						
OSL	In-Air	Distance Test	80	0.1	11/02/2024	
OSL	In-Water	Distance Test	50	2.0	12/02/2024	
HW Loch	In-Water	Distance Test	40	52.97 68.02 112.69 142.31	12/02/2024	
b'RELEASE' Command Testing						
OSL	In-Air	Send/Receive Data	80	0.5	12/02/2024	
HW Loch	In-Water	Send/Receive Data	40	52.97 68.02 112.69 142.31	12/02/2024	Conducted at same time as Distance Test (12/02/2024)
OSL	In-Air	Send Data from laptop to modem configured with release mechanism	80	0.5	12/03/2024	
OSL	In-Water	Send Data from laptop to modem configured with release mechanism	60	3.0	12/03/2024	
OSL	In-Air	Send Data from App to modem configured with release mechanism	100	0.5	21/03/2024	
OSL	In-Water	Send Data from App to modem configured with release mechanism	75	3.0	21/03/2024	

Table 8 Testing Results

4.2 Demonstration

A demonstration was carried out within the Ocean Systems Lab at Heriot Watt University on 27/03/2024.



Figure 35 Demonstration Video

4.3 Mechanical Testing and Results:

4.3.1 Testing materials for endcaps

4.3.1.1 Method

Before adding the endcap, it was important to test the different materials. A successful material is one which is able to conduct the magnetic field from one side of the material where the magnet is located, to the opposite side of the material where the stainless-steel disc is then connected to the rope and the buoy. Conducting the magnetic field of the magnet through the endcap to the stainless-steel disc is paramount, as it ensures that it can hold down the buoy even in the strong underwater currents of the sea. Additionally, the stronger the magnetic field, the bigger the buoy that can be used, making it easier for fishermen to locate and retrieve their traps.

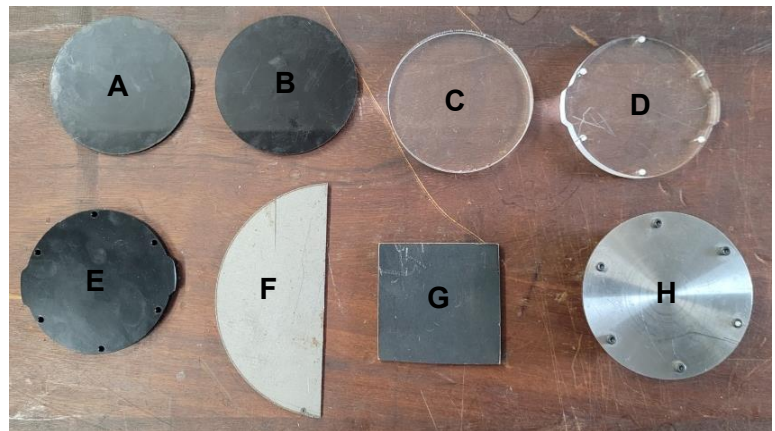


Figure 36 Different materials used.



Figure 37 Test rig

The different materials tested were a range of acrylic pieces of different thicknesses, as well as 2 types of mild steel and stainless steel. These are shown in Figure 36.

To test the different materials reliably, a simple test rig was put together (Figure 37) consisting of a clamp stand to hold the enclosure, a plastic bucket, and scales.

While the magnet was switched to the “on” position, each material was placed between the magnet and the stainless-steel disc. The stainless-steel disc was attached to the plastic bucket. Water was slowly added to the plastic bucket until the bucket detached due to the excessive mass. The mass of the plastic bucket, with the stainless-steel disc and the water was then measured using scales and noted down in a table. This process was repeated three times for each material for all 8 materials.

4.3.1.2 Results

Table 9 shows the results and the average mass that each material could hold.

			Mass(kg)			
			1st Attempt	2nd Attempt	3rd Attempt	Average
Material	A	3mm Acrylic (Black)	1.751	1.854	1.754	1.786
	B	5mm Acrylic (Black)	1.038	0.987	1.002	1.009
	C	10mm Acrylic (Clear)	0.143	0.168	0.151	0.154
	D	9mm Acrylic (Clear BlueRobotics)	0.204	0.189	0.191	0.195
	E	6mm Aluminium (Black BlueRobotics)	0.748	0.678	0.802	0.743
	F	2mm mild steel	5.184	4.987	4.905	5.025
	G	6mm mild steel	0.109	0.158	0.128	0.132
	H	3mm Stainless Steel	4.371	4.293	4.352	4.339

Table 9 Magnet Strength Testing

4.3.1.3 Analysis

By analysing the table, it can be seen that metal is clearly better at conducting magnet fields than acrylic. The material which performed the best was the 2mm stainless steel, which held an average of 5.025 kg before detaching. The seconds best material was the 3mm stainless steel which successfully held an average mass of 4.339 kg.

4.3.1.4 Conclusion

Although the 2mm mild steel is stronger than the stainless steel, it was decided that the stainless steel was still the better option for two reasons. The main reason being that mild steel will corrode more in seawater, which will greatly reduce its structural integrity, whereas stainless steel will not corrode as much. Another reason for using the stainless steel opposed to the mild steel is that the mild steel was just too thin to make a strong endcap that would not implode.

It was also observed that the magnet the steel disc could hold more weight if the surface between it and the stainless-steel endcap was dry.

5 Conclusions and Future Work

5.1 Conclusion

The ropeless fishing solution fulfilled the key requirements of the project and the progress made exceeded the initial expectation of what could be achieved in the short time period. A strong prototype was created and tested in a water tank to simulate the true operating conditions which was a major extension on previous work. The communication process, critical to the project, was improved by streamlining the operation of the Nanomodems and introducing Bluetooth Low Energy to the mobile app. In addition to this, the buoy release mechanism was completely revamped enabling reliable set and release operations to be performed.

The solution offers a unique alternative to existing solutions on the market at a considerably lower price. The testing of the prototype was successful in proving the feasibility of the solution, but it also highlighted some areas of potential improvement which are outlined below. Ultimately, if the ropeless fishing solution prototype is to be continued for productisation, there are some additional features that if implemented could greatly improve the operation and adoption of the product.

5.2 Future Works

5.2.1 Future Works Overview

Building upon the successful prototype, several aspects were identified that could further enhance the system. This section underlines the different ways in which the product could be further developed through the implementation of additional features which may reduce cost and risk of corrosion while increasing safety and battery life, and improving usability.

5.2.2 Pricing

Below, Table 10, demonstrates the per unit pricing of the overall system currently.

Seabed Unit		
2	Flange	70\$
1	4-hole Endcap	14\$
1	240mm x 75mm Enclosure	180\$
1	Nanomodem 3	100\$
1	Raspberry Pi Pico W	13\$
1	Servo	20\$
1	Magnet	32\$
1	Steel Endcap	13\$
1	Vent	28\$
2	Closed penetrator Kit	6\$
1	Battery Pack	10\$
Total		486\$ / £382

Surface Unit		
2	Flange	70\$
1	4-hole Endcap	14\$
1	150mm x 75mm Enclosure	150\$
1	Nanomodem 3	100\$
1	Raspberry Pi Pico W	13\$
1	Aluminium Endcap	14\$
1	Vent	28\$
2	Closed penetrator Kit	6\$
1	Battery Pack	10\$
Total		405\$ / £ 318

Table 10 Unit Pricing

5.2.3 Communications Future Works

As part of the ongoing efforts to enhance the efficiency and sustainability of the Ropeless Fishing Gear system, future developments to address power management and system reliability are required. These enhancements aim to extend the operational lifespan of the devices and ensure consistent performance under varying environmental conditions.

Low Battery Mode:

A significant enhancement is the introduction of a Low Battery Mode for the seabed unit. In this mode, the modems will automatically switch to a minimal power usage state when instructed by the surface unit. The mode can be activated via the mobile app or programmed to trigger based on a schedule that suits seasonal fishing activities or specific fishing operations. For example, during off-peak seasons or known periods of inactivity, the modems can enter a dormant state for extended periods, effectively shutting down all non-essential functions. This would drastically reduce power consumption, thereby preserving battery life when the system is not needed.

Failsafe for Low Battery:

To prevent system failures due to depleted batteries, a robust failsafe mechanism could be implemented. This mechanism would monitor battery levels continuously and initiate a series of actions if the power drops below a predefined threshold. When the battery level is critically low, the unit could automatically send a notification to the user's mobile app, advising them of the need to retrieve the gear or replace the batteries. Additionally, if no action is taken and the battery continues to drain, the system would automatically release the buoy mechanism and allow for retrieval.

5.2.4 Mechanical Future Works

Stainless steel has been used for the endcap and disc due its great magnetic conductivity and being mostly corrosion proof. Since the sea is a very harsh environment, no metal is truly safe from corrosion, which has led engineers to come up with different ways of solving this problem, the most common and successful being "cathodic protection". Cathodic protection is a technique used to prevent corrosion of metal surfaces by making them the cathode of an electrochemical cell. This is typically achieved by applying a sacrificial anode or by impressing an external current onto the metal surface, effectively slowing or stopping the corrosion process. This would be extremely useful for the developed product.

If the product is to be commercialised and used by fishermen and marine scientists alike, it is crucial to reduce the total cost of the system as much as possible while maintaining high quality standards. This can be achieved in a variety of ways, but perhaps the most important one is to reduce material costs. By mass producing components such as the enclosure, flanges, endcaps, and magnets, this would decrease the price of the overall system substantially.

One mechanical aspect which has not been thoroughly explored in this project yet is the rope coiling system. This system consists of a long rope connecting the enclosure to the buoy and the disc. For it to be used in the ocean, a bag must be used to store the length of rope required (~300m). This rope must be able to fit inside a small bag and be released on demand without becoming tangled.

5.2.5 Electronics Future Works

One of the requirements for the PCB is that its size is reduced as much as possible so it can properly fit in the pressure vessel. Actions were taken to make this possible, including the removal of the DB9 connector and the optimisation of the PCB design. Despite these efforts, the new PCB's dimensions are still not ideal and can still be minimised even more, even though it does fit in the current pressure vessel. By using a different PCB assembly technique, the height of the PCB could still be lowered. It is not certain if the

Electronics Workshop would be able to accomplish this, so perhaps using an external manufacturer would be a better option.

The addition of other sensors such as pressure, temperature, and leak sensors as well as a camera would greatly benefit the product. Pressure sensors and temperature sensors could provide valuable data to marine scientists responsible for tracking the conditions of seawater. A leak sensor could provide feedback to the user if a leak has been detected, prompting the user to collect the enclosure before it becomes permanently damaged. Additionally, the inclusion of a camera could also benefit both fishermen and scientists as the fishermen would be able to observe the effect of different bait on the crustaceans. Marine scientists could also benefit from camera data, as they would be able to observe fish behaviour.

5.2.6 Mobile App Future Works

The mobile app fulfils the main requirements of the project and successfully handles the communication between the mobile device and surface level Nanomodem. However, one potential improvement would be to pin the latitude and longitude values collected for each trap during the setup process to a map within the app. This feature would allow fishermen to visualise the location of their traps and easily navigate between them. The map feature could be implemented using existing programming libraries based on Google maps which would simplify the integration process and take advantage of existing well-established tools.

A second feature that could improve the usability of the mobile app, would be allowing users to enter a duration of time or a specific time to be reminded of the status of their trap. This would help fisherman manage their traps as they could be notified when a trap approaches a desired time of release. It would also add a layer of protection against fishermen forgetting to collect their trap within the maximum time period which would be dictated by the battery life of the receiving Nanomodem on the sea floor.

A final feature that would benefit the app would be by providing an acknowledgement when a command sent from the app has been successfully received by the Nanomodem on the sea floor. This would reassure users that their command was successfully received and that they can expect the buoy associated with the trap they are releasing to rise to the surface. This feature has not yet been implemented as the Nanomodem on the sea floor would have to send an acknowledgement signal to the Nanomodem at the surface which would finally relay this signal to the mobile app. Due to the two-step communication process, sending an acknowledgement back to the mobile app is a complex process and one that impacts the battery life of the Nanomodem on the sea floor. To successfully implement this feature, it would be important to streamline the acknowledgement communication process and perform further testing on it to ensure its reliability. In addition to this, calculations should be performed to determine how significantly sending the acknowledgement signal would affect the battery life of the Nanomodem and these calculated values should be factored into the maximum deployment time. At the moment, the mobile app the app utilises an intermediary solution which requires a user to confirm the success of the operations they have requested, reducing the importance of an automatic acknowledgement.

5.2.7 Future Works Conclusion

The most crucial stage of any product after production is Testing. Testing provides quality assurance, safety, customer satisfaction, compliance, reliability, and durability as well as identifying further improvements. The product would greatly benefit from more testing. This testing should be conducted in different conditions including seawater and freshwater environments. The tests should be performed with Nanomodems at different distances from each other, vertically and horizontally, to establish the maximum range of communication. Factors such as number of successful and unsuccessful attempts, time taken for the buoy to float up to the surface at different depths must also be tracked as they provide reliability and durability data.

6 References

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7 Appendix

7.1 Product Specification

1. Performance under Specified Conditions

- 1.1. The Ropeless Fishing Gear Demonstrator (RFGD) is designed to function within the epipelagic zone of the submarine environment, capable of idling for 20 hours and continuously monitoring for activation commands for a 4-hour window, defined by the user. Upon receiving the specific command, it activates its release mechanism to turn the servo, subsequently deactivating the magnet and releasing the buoy to the surface for retrieval purposes.
- 1.2. The release system will be triggered with the command "*b'RELEASE*" or at 408 hours.
- 1.3. The release mechanism will not exceed an operating speed of 0.18s/60° @ 5V.
- 1.4. Servo motor will have a maximum torque of 14kg/cm @ 5V.
- 1.5. Release mechanism will have a range of rotation of 90°.
- 1.6. Nanomodem v3.2 acoustic modem range of 2km.
- 1.7. RFGD will have a response time of <1s.
- 1.8. The device uses an acoustic frequency range of 24⇌32kHz.

2. Environment

- 2.1. Submarine depths of <200m.
- 2.2. Ambient operational temperature range of
5°C ⇌ 40°C ± 5°C *charging*
-15°C ⇌ 55°C ± 5°C *discharging*

3. Design

- 3.1. Must detect command and activate release mechanism.
- 3.2. Must release buoy on command within specified window.
- 3.3. Must autoactivate release mechanism if no command is received within 17 days (408hours).

4. Power

- 4.1. The product should be powered from a 5v DC Voltage, 2-amp source.
- 4.2. The RFGD has an estimated battery life of 504 hours (21d or 3w).

5. External Body

- 5.1. The external body of the product is constructed from either:
Clear acrylic rated to depths of 150m.
Black aluminium rated to depths of 950m.
- 5.2. The product will consist of an internal structure, for each of the components. These will provide a secure housing for internal mechanisms and devices.
- 5.3. Internal structure is constructed from PLA (Poly-lactic Acid).

6. Period of Useful Life

- 6.1. The RFGD has an estimated working life of 500 cycles before requiring any maintenance or servicing.
- 6.2. A cycle is defined as discharging from a complete charge to autoactivated release mechanism -
100% ⇌ ~30%

7. Total Life

- 7.1. The RFGD should last in excess of 1000 Cycles with nominal maintenance / servicing before becoming completely inoperable.

8. Safety

- 8.1. Rope attachment for the buoy should be stored to minimise the risk of entanglement or injury to marine life when deployed.
- 8.2. Servo motor should be fitted with overload protection in the event of an obstacle.
- 8.3. The system should not be deployed at depths beyond the rated limit for the chosen enclosure.
- 8.4. Ensure the system is deployed within the maximum length limits of the rope/buoy to avoid exceeding its operational depth range.
- 8.5. The system should not be deployed at depths beyond the range of acoustic modem.

9. Warranty

- 9.1. No warranty is offered on any aspect of the product.

10. Marketing

- 10.1. Summary of market requirements:
 - 10.1.1. Minimise environmental impact, protecting marine life and habitats.
 - 10.1.2. Enable trap retrieval without surface markers.
 - 10.1.3. Ensure resistance to saltwater, pressure, and weather.
 - 10.1.4. Offer an intuitive system for gear tracking and retrieval.
 - 10.1.5. Meet local and international fishing and conservation standards.
 - 10.1.6. Use of rechargeable power source.

11. Packaging

- 11.1. Packaging / transport cost should be kept to a minimum and preferably below 5% of the unit cost.

12. Size and Weight Restrictions

- 12.1. Enclosure
 - 12.1.1. Weight to not exceed 5kg.
 - 12.1.2. Length of enclosure to not exceed 300 mm.
 - 12.1.3. Diameter of enclosure to not exceed 100mm.

13. Shipping

- 13.1. No shipping is offered for the product.

14. Manufacturing Processes

- 14.1. Manufacturing process will commence upon request.

15. Motors, Sensors, and Controllers are bought in from the following suppliers:

- 15.1. Motors:
 - 15.1.1. FEETECH High-Torque Servo FS5115M – RobotShop
- 15.2. Controller:
 - 15.2.1. Raspberry Pi Pico W – RS Components
- 15.3. Sensor:
 - 15.3.1. Nanomodem V3.2 – Newcastle University
- 15.4. 3D Filament
 - 15.4.1. PLA 3D printer Filament – RS Components
- 15.5. Power:
 - 15.5.1. 5V, 2A PSU – EnergyCell
- 15.6. PCB
 - 15.6.1. PCB manufacture – Heriot-Watt Electronics Workshop
- 15.7. Magnet
 - 15.7.1. 80kg Switchable Magnet - Neoteck

16. Aesthetics

- 16.1. The form will follow function.
- 16.2. If cost dictates, the RFGD should look attractive/ to improve market perception with a focus on the enclosure and release mechanism.

17. Ergonomics

- 17.1. Mild Steel handle placed on length of enclosure for ease of use.
- 17.2. The product should be unmanned unless maintenance is required.

18. Customer Requirements

- 18.1. See *Marketing*

19. Competition

- 19.1. The RFGD will be operating against equivalent models which include the following companies:
 - 19.1.1. Sustainable Seas Technology, Inc – Connecticut, USA
 - 19.1.2. Desert Star Systems LLC – California, USA

20. Quality and Reliability

- 20.1. Quality should be such that actuators and sensors should not generally fail within a period of three years and only <1% should fail within the first year.
- 20.2. The Servo motor should not fail in the area of overload.

21. Standards and Specifications

- 21.1. Standards to be adhered to:
 - 21.1.1. BS 5000, part 99 motor performance
 - 21.1.2. BS 2754 electrical insulation
 - 21.1.3. BS 7671 Requirements for electrical installations. IET wiring regulations.

22. Processes

- 22.1. All components to be of metric form and comply with ISO 9001 for quality management.

23. Testing

- 23.1. Testing is to be carried out on 5% of units.
- 23.2. All cables should be tested to BS7671.

24. Legal

- 24.1. Possible litigation lies in the user injuring themselves by having access to moving parts during operation.

25. Documentation

- 25.1. Product should be supplied with a user manual covering RFGD operation and maintenance.
- 25.2. Suppliers require maintenance and repair manual.

26. Disposal

- 26.1. PLA parts should be separable and modular to aid disposal.



Ropeless Fishing Gear Demonstrator

Group 2

Dr Keith Brown

Problem with trapped Fisheries



- Trap Rope Entanglement
- Theft
- Marine Ecosystem and Biodiversity Disruption



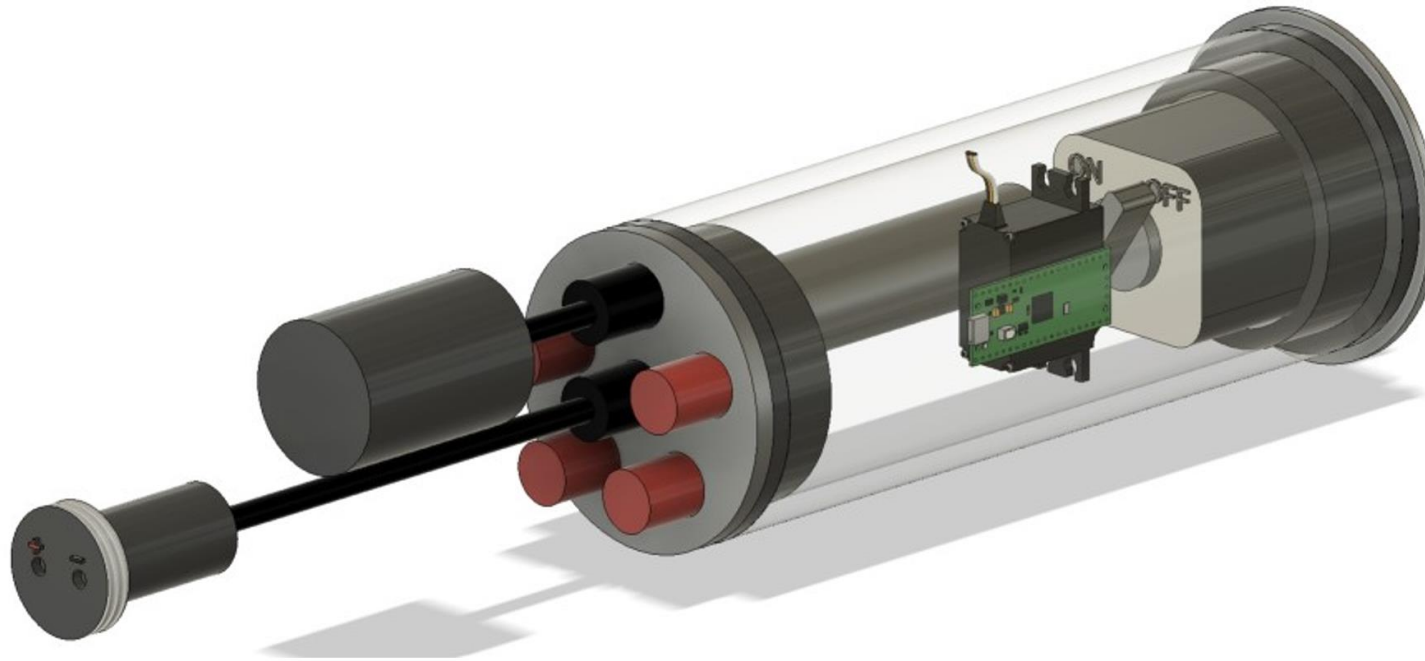
Project Goals



- ① Develop a Cost-Effective Prototype
- ② Ensure Reliable Release Mechanisms
- ③ Create User-Friendly Software for Tracking



Overall Design



Challenges - Magnetism



- Conducting magnetism through different materials
- Reliably securing a servo motor to turn the magnet
- Fitting the magnet in the enclosure

Challenges - Waterproofing



- Flanges
- O-rings
- Penetrator kits
- O-Rings
- Molykote lubricant



Challenges – Nanomodem v3.2



- Non-Standard Wiring
- UART to RS232 Protocol
- Multipath Propagation

Challenges - Power



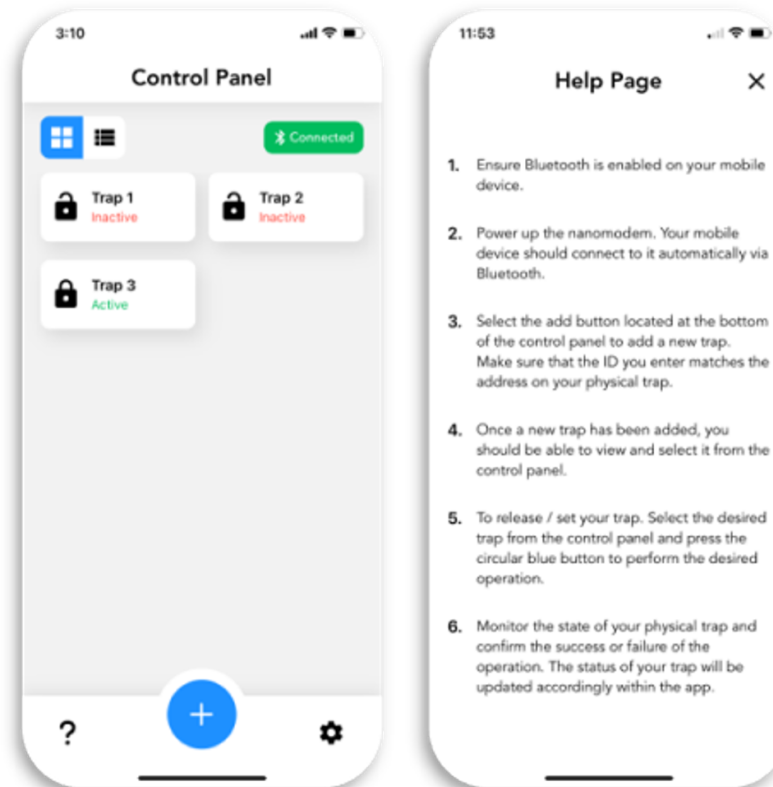
- Power Consumption
- Battery Capacity
- Raspberry Pi Pico W Deep Sleep



Challenges – User Application



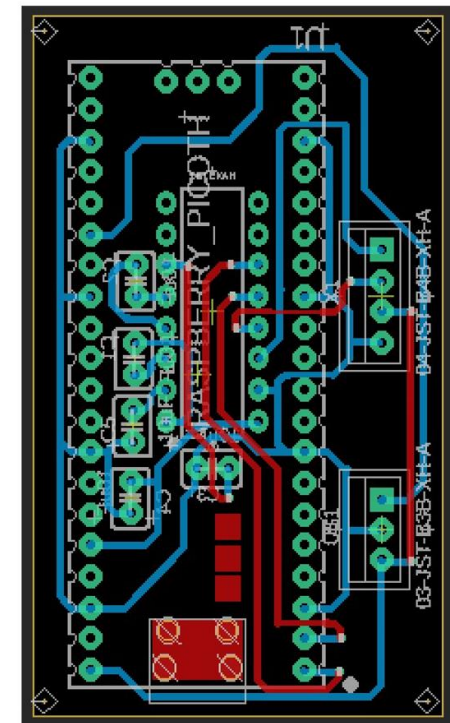
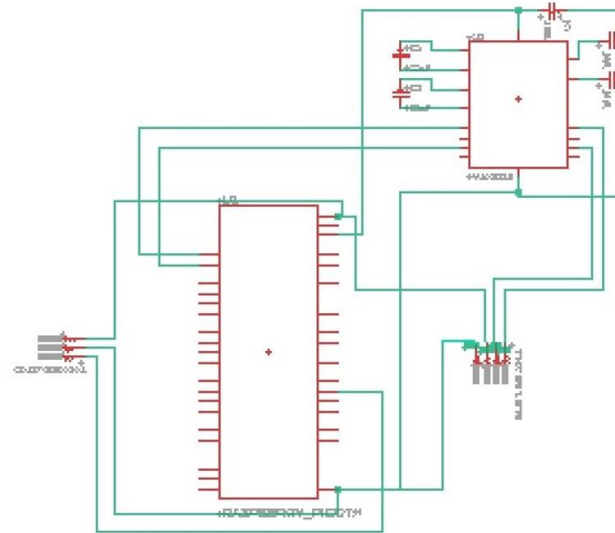
- Poor user adoption due to app complexities
- Maintaining reliable communication with nano -modem
- Power consumption
- Scalability



Challenges – Printed Circuit Board



- UART ↔ Rs-232
- Servo Control
- Size Management



Results



Product Specification



- Created provisional Product Specification
- Current state of the system
- Demonstrate current limitations & achievements

Outcome



- Successfully developed a product which meets the goal
- Conducted a variety of tests to ensure product reliability
- Built a product which may be later commercialised

Conclusion/Recommendations



- ① Reduce form factor
- ② Enhance PCB for scalability
- ③ Develop App functionality